

MOHAMMAD ABRARUL HASANAT

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RESEARCH INTEREST

Distributed Machine Learning, Databases, Systems for Machine Learning and Software Engineering, Systems, Distributed System.

EDUCATION

Bangladesh University of Engineering and Technology (BUET)

Bachelor of Science in Computer Science and Engineering

Dhaka, Bangladesh

Apr 2019 – July 2024

- **CGPA:** 3.72/4.00

RESEARCH EXPERIENCE

Multi Cluster Federated Learning based on Statistical Similarity

Dec 2023 - Present

Supervisor: Dr. Muhammad Abdullah Adnan (BUET)

Undergraduate Thesis

- The object of this research is to reduce communication cost and improve accuracy in Horizontal and Vertical Federated Learning
- In this research, we propose a Multi-Clustered Federated Learning framework with the help of Principal Component Analysis. We have so far tested on vertically split data to group them into multiple clusters based on their statistical properties (PCA) and generate multiple models. Tests on several benchmark datasets achieved satisfying results compared to other existing approaches. Now we are aiming to improve communication cost and accuracy in both Horizontal and Vertical approaches.

Distributed Machine Learning

2023

- I investigated Distributed Machine Learning using Parameter Server and various communication policies like ASP, SSP, and BSP. I also acquired hands-on experience in dynamically adjusting buffer size and synchronization switching to optimize performance
- I have used Docker Containers for running Distributed machine learning programs. We simulated a distributed system by using multiple Docker containers that were ssh able from each other. My understanding of cloud platforms also extends to using cloud storage solutions and integrating cloud services with application development. My overall expertise is on building systems and scheduling tasks in ML and FL tasks and reducing heterogeneity across clients and data.

PROFESSIONAL EXPERIENCE

Software Development Engineer1

July 2024 - Present

IQVIA (Fortune 500)

- Collaborating to frontend development by implementing new features, resolving bugs and testing it in predev environment while collaborating efficiently across teams and taking help from senior expertise.
- Applying SWE principles and version control skills resulting in efficient and maintainable code contribution.

ACADEMIC PROJECTS

TourCell

2023

- We developed a tour management project where vendors can offer tour packages along with multiple flights and accommodation options and a customer can filter out desirable tour packages and customize them based on their needs. It also includes emergency notification and live tracking of tour attendees.
- **Stack:** React Js, Spring Boot, Redux, PostgreSQL
- Front-End, Back-End

A Simple C-Compiler

2023

- A simple compiler for the language described in the book "Compilers: Principles, Techniques, and Tools" by Aho, Sethi, and Ullman. It is not a complete compiler, but it does implement symbol table, a lexer, a parser, and a code generator
- **Languages:** C++, YACC, Lex, Shell

- [Github](#)

VaccineForAll

2022

- Developed a flexible COVID-19 vaccination management system with separate user and center site. It allows user to take Vaccines from different centers and different types of vaccines. Centers can set appointment, update vaccine storage, vaccination status of a person and verification
- **Stack:** React Js, Express JS.
- [Github](#)

Ray Tracer in C++

2023

- Developed a ray tracer with recursive reflections, lighting (ambient, diffuse, specular), and texture mapping. Implemented ray-object intersections for planes, spheres, and cubes in a 3D OpenGL scene. It also supports dynamic camera controls, shadows and reflections.
- **Languages:** C++
- [Github](#)

TECHNICAL SKILL

- **Programming Language:** C, C++, Python, Java, JavaScript, SQL, Bash, Intel 8086 Assembly Language.
- **Frameworks:** Angular, React, Node.js, JavaFX, Spring Boot.
- **Database:** Oracle, MySQL, PostgreSQL.
- **Version Control:** Git.
- **Misc:** Docker, Kubernetes, AWS.

CONTEST EXPERIENCE

- 31st place in ICPC 2021-Asia Dhaka Preliminary Round [view certificate](#)
- Honorable place in 2022 and 2023 ICPC Asia Dhaka Preliminary Round [view certificate](#)
- Over 1500+ problems related Data Structure and Algorithm related Problems solved in different online judges
- Codeforces Max Rating 1717. (Current Rating 1689)

References

- **Dr. Muhammad Abdullah Adnan**
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- **Dr. Rifat Shahriyar**
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